Original-equation certification and quotient observables in VigilODE: theory-first integration of the RVJ/NAG research and a bounded native validation

VigilODE research and validation record
(Dated: October 28, 2026)

We critically integrate the latest cumulative VigilODE validation, the supplied RVJ/NAG research archive, and conditional earlier polyalgorithm results. Three independent theoretical tasks precede parent admission and three coding tasks. For a fixed cyclic algebraic closure we prove sharp Bernstein weights 3/4 and 25/32, a safe exponential-free physical-error recurrence, and an endpoint reuse identity. A scalar $O(n)$ lift avoids runtime Jacobian assembly, matrix factorization, Newton iteration and derivative callbacks. For an explicit quartic quotient, a multiplicity-preserving trace remains defined through a triple root and admits a shared polynomial field; its semantics cannot replace a selected physical root. The published execution source passes 15 native unit tests, 732,947 independent exact-rational checks covering every exported slab, and 10 supporting symbolic identities. At the same exact whole-state error cap 10^{-8}, the first passing registered cyclic grid changes from 64 to 32 steps; native whole-call median speed ratios are 8.72–9.33 in the measured cohort. Trace field kernels improve, but the complete candidate-trajectory measurements are mixed and do not establish a speedup. We retain rejected grids, inherited negative results, unresolved singular-branch and multivariate-quotient questions, and all cumulative qualification gates. The general production status is HOLD.

I. SCOPE, EVIDENCE AND THE LATEST-PR AUDIT

The execution source is

`64004e23042f2cf076da5378be724f50c3856f4c`.

It was published and remotely read back before the scientific run. Its source tree is `67564b455011252fdc1fb811c09baeea9f2bd31f`. The parent is PR96, commit `13f3298bd0213f121e07385903f9508c435ad6a1`. PR95, commit `36104bfc65d1fdc695c8acea383c54b50a1c4e78`, was read as a separate theoretical input and was not merged. The delivery branch is `research/production-rvj-nag-20261028`. The supplied JSON and ZIP, and 23 byte-preserved selected source files, are bound by the intake manifest. Historical measurements are inputs to criticism; they are not the baseline for cross-host or cross-language speed ratios.

PR96 contains useful scoped repairs: an exact-rational squared residual check for the returned Givens solution and a corrupted-answer control, explicit entered/refused/completed/failed callback accounting, and focused lint evidence. We verified 24 compact evidence hashes and five selected source hashes. The measured-to-final PR96 code difference in the relevant runtime/tool paths is empty. These facts do not complete its remaining production gates. The prior authority check had 49 unresolved references, Q-API remained HOLD, the six N tasks were unfinished, and all ten interaction gates were NOT_RUN. The W-filtered negative cost ratio 15/7 exceeded the declared 1.3 threshold; the inaccurate Off result is retained. An earlier timeout and its extended test run overlap and must not be added as independent test counts.

Some older raw files and an exact Givens receipt were left in unavailable temporary directories; compact summaries are not fresh replays. PR96 also distinguished its preregistration commit from a later actual execution source that had not been remotely published before execution. The present cohort closes that provenance gap for its own source only. Existing formatting, ignored-feature, vendor, public-API and external-client obligations remain. The final unmodified authority check first timed out during lazy object retrieval. A cached-object run with lazy fetch disabled failed with 178 distinct unallowlisted missing objects and 919 failure citations. This partial-clone result is not directly comparable with the earlier fuller-checkout count of 49; missing objects can include blobs. Three new-node citations refer to the byte-preserved external BASS commit/blob input, which the untyped repository checker treats as VigilODE authority. No allowlist was expanded. The complete audit and cumulative correction counters are delivered separately in `theory/PR96_AUDIT.md`, `theory/PRODUCTION_GATES.json`, and `LOCAL_DEVELOPMENT_PLAN.json`.

A. Research order and admissible claims

The theoretical agents investigated (i) original-equation cyclic certificates, (ii) trace algebra and branch semantics, and (iii) PR96 evidence and applicability. The parent reviewed those results and issued a bounded theory admission before runtime implementation. The coding phase separated the two native modules from the independent exact

oracle, followed by an adversarial review. One coherent prepublication correction episode is charged to inherited N03 and N05, each 1/3 at scientific execution. A subsequent CI rustdoc-only correction charges N03 a second episode; N03 is now 2/3 and N05 remains 1/3. Four mathematical interval literals were wrapped as code. Executable source lines, constants and the oracle are unchanged; no numerical campaign was rerun. R-STRUCT and R-ADAPT remain exhausted at 3/3; Q-LEDGER remains 1/3.

The public modules are opt-in structured components, rather than a completed generic stiff solver. A polyalgorithm may choose between certified candidates only after their original model, requested observable, physical norm, clock, checkpoint and output schedule agree. The cyclic small-real branch and the all-complex multiplicity trace are different targets. A shared cost model is meaningful only within a common admissible target.

II. ORIGINAL-EQUATION CERTIFICATES FOR THE CYCLIC CLOSURE

The cyclic problem has indices modulo n and original equations

$$G_i(x,q) = q_i + \frac{1}{4}q_{i+1}^2 - x_i = 0, \tag{1}$$

$$\dot{x}_i = f_i(x,q) = -x_i + \frac{q_i + q_{i+1}}{16} + c_i, \qquad c_i = \frac{1}{32} + \frac{i \bmod 3}{256}. \tag{2}$$

The declared branch starts at $x = q = 0$, with physical time $0 \le t \le 1$, and domains $X = [-1/8, 1/8]^n$, $Q = [-1/4, 1/4]^n$. The native specialization admits $n = 4, 8, 16$. The required error is the supremum over the entire time interval and all slow and algebraic coordinates, in the original infinity norm, with cap $10^{-8} = 1/100000000$.

a. Proposition C1: branch and physical-error transfer. There is a unique $q(x) \in Q$ for each $x \in X$. Moreover,

$$\|q(x) - q(z)\|_\infty \le \frac{8}{7}\|x - z\|_\infty, \tag{3}$$

$$\|\widehat{q} - q(x)\|_\infty \le \frac{8}{7}\|G(x,\widehat{q})\|_\infty \quad (\widehat{q} \in Q). \tag{4}$$

The reduced field $F(x) = f(x, q(x))$ contracts at rate $\lambda = 6/7$.

Proof. The map $T_x(q)_i = x_i - q_{i+1}^2/4$ maps Q into $[-9/64, 1/8]^n \subset Q$ and contracts with factor $1/8$. Subtracting its fixed-point equations, or comparing $\widehat{q}$ to its fixed point, gives the two bounds. On an upper face of X, $f_i \le -7/128$; on a lower face, $f_i \ge 1/8$. Thus the exact slow solution stays in X. Writing $F = -x + Cq + c$ with $\|C\|_\infty = 1/8$ gives the maximum-coordinate Dini contraction rate $1 - (1/8)(8/7) = 6/7$.

For a differentiable candidate $(\widehat{X}, \widehat{Q})$ in $X \times Q$, define the original residuals

$$R_f = \widehat{X}' - f(\widehat{X}, \widehat{Q}), \qquad R_g = G(\widehat{X}, \widehat{Q}). \tag{5}$$

Whole-slab bounds $\|R_f\|_\infty \le D_f$ and $\|R_g\|_\infty \le D_g$, with $e = \|\widehat{X} - x_{\text{exact}}\|_\infty$, imply

$$D^+ e \le -\lambda e + \delta, \qquad \delta = D_f + D_g/7, \tag{6}$$

$$\|\widehat{Q} - q(x_{\text{exact}})\|_\infty \le \frac{8}{7}(e + D_g). \tag{7}$$

This comparison uses a residual evaluated on the numerical reconstruction. It therefore retains the exact-flow error anchor; an exact-input-only discrete truncation estimate cannot be substituted without a separate stability argument. Adjoint DAE analysis for linear quantities of interest provides complementary output contracts [3]; the present contract retains every physical coordinate throughout the path.

A. Finite lift and the exact reconstructed path

Let $b_i = q_{i+1}/2$, $(Av)_i = b_i v_{i+1}$ and $P = \prod_i b_i$. Then $A^n = PI$ and

$$(I + A)\sum_{k=0}^{n-1}(-A)^k = [1 - (-1)^n P]I. \tag{8}$$

The denominator $d = 1-(-1)^n P$ satisfies $d \geq 1-8^{-n} > 0$ on Q. Solving $v_i = f_i - b_i v_{i+1}$ by composition around the cycle and a backward substitution computes the inverse action in $O(n)$ scalar operations. For the exact rational lift $s(y) = (f, v)$, $y = (x, q)$,

$$G_q v = f \tag{9}$$

holds even at endpoints with $G \neq 0$. The native circuit evaluates this declared analytic structure without a Jacobian, JVP, AD or Newton callback. The identity is specific to these equations and is not a black-box result for arbitrary closures.

Stored finite binary64 endpoints y_0, y_1 denote exact dyadics. The certified Hermite polynomial uses the *exact rational* slopes $s_j = s(y_j)$, enclosed numerically by outward arithmetic. With $B_j^m(\theta) = \binom{m}{j}\theta^j(1-\theta)^{m-j}$, its cubic Bernstein controls are

$$Y_0 = y_0, \quad Y_1 = y_0 + hs_0/3, \quad Y_2 = y_1 - hs_1/3, \quad Y_3 = y_1. \tag{10}$$

The componentwise control hull must lie in $X \times Q$. The Bernstein convex-hull property provides the whole-interval range enclosure [1]. Arithmetic enclosure remains necessary: rounded endpoint slopes would define a different path. The supplied dense-query interface returns an interval enclosure of the declared path.

a. Proposition C2: sharp slow-residual weight. For one slow component let X_j denote its controls and set

$$D_0 = f_i(y_0), \qquad D_2 = f_i(y_1), \tag{11}$$
$$D_1 = 3(X_2 - X_1)/h, \tag{12}$$
$$r_1 = D_0/3 + 2D_1/3 - f_i(Y_1), \tag{13}$$
$$r_2 = 2D_1/3 + D_2/3 - f_i(Y_2). \tag{14}$$

The cubic Bernstein residual coefficients are exactly $(0, r_1, r_2, 0)$, and consequently

$$D_f \leq \frac{3}{4}\max_i\{|r_{1,i}|, |r_{2,i}|\}. \tag{15}$$

Proof. Degree elevation of the quadratic derivative gives the displayed interior coefficients. Affineness of f gives field controls $f(Y_j)$; the endpoint differences vanish by the exact-slope definition. The residual equals $3\theta(1-\theta)[(1-\theta)r_1 + \theta r_2]$. Since $3\theta(1-\theta) \leq 3/4$, Eq. (15) follows, with equality possible for equal, same-signed r_1, r_2 at $\theta = 1/2$.

b. Proposition C3: three interior algebraic coefficients. For component i set $d_j = Q_{i,j} - X_{i,j}$, $z_j = Q_{i+1,j}$ and $g_\ell = G_i(y_\ell)$. The degree-six residual coefficients are

$$(g_0, g_0, \beta_2, \beta_3, \beta_4, g_1, g_1), \qquad \beta_k = E_k + S_k/4, \tag{16}$$

where the exact rational expressions are

$$E_2 = (d_0 + 3d_1 + d_2)/5, \qquad S_2 = (2z_0z_2 + 3z_1^2)/5, \tag{17}$$
$$E_3 = (d_0 + 9d_1 + 9d_2 + d_3)/20, \qquad S_3 = (z_0z_3 + 9z_1z_2)/10, \tag{18}$$
$$E_4 = (d_1 + 3d_2 + d_3)/5, \qquad S_4 = (2z_1z_3 + 3z_2^2)/5. \tag{19}$$

Proof. Cubic-to-degree-six elevation gives E_k and the Bernstein product rule gives S_k. Equation (9) makes the two endpoint derivatives of $G(Y(\theta))$ zero. The derivative formulas $6(\beta_1 - \beta_0)$ and $6(\beta_6 - \beta_5)$ then give the repeated endpoint coefficients, without assuming exact constraint satisfaction.

Let g bound the endpoint magnitudes and m the three interior magnitudes, over all components. The interior basis weight is

$$w = B_2^6 + B_3^6 + B_4^6 = 15u^2 - 10u^3, \qquad u = \theta(1-\theta) \in [0, 1/4]. \tag{20}$$

It increases with u and attains $25/32$. Hence

$$D_g \leq \max\{g, (7g + 25m)/32\}. \tag{21}$$

The weight is sharp for this coefficient-information class. All these constants follow from exact polynomial identities; none is fitted.

c. Proposition C4: endpoint reuse. The same represented endpoint and model/arithmetic identity define the same exact slope and original residual. Thus N consecutive slabs need $N+1$ endpoint preparations instead of $2N$. The native reuse check requires every endpoint bit, dimension and model epoch to agree; branch, domain and arithmetic provider are fixed by the module. For a nonautonomous extension the physical time would also belong to this identity. A streaming consumer needs only the adjacent prepared endpoints. This is an exact work reduction, with runtime effects measured separately.

B. A safe exponential-free error recurrence

a. Proposition C5: maximum of two rational envelopes. Let $E, \delta, h \geq 0$, $\lambda > 0$ and $z = \lambda h$. Define

$$U_1 = \frac{E + h\delta}{1+z}, \qquad U_2 = \frac{(1-z/2)E + h\delta}{1+z/2}, \tag{22}$$

$$E_+ = \max(U_1, U_2), \qquad E_{\text{whole}} = \max(E, E_+). \tag{23}$$

These enclose the endpoint and whole-slab comparison solution of Eq. (6).

Proof. Writing $w = \delta/\lambda$, the exact comparison is $T(h) = w + (E-w)e^{-z}$. For $z \geq 0$,

$$\frac{1-z/2}{1+z/2} \leq e^{-z} \leq \frac{1}{1+z}. \tag{24}$$

The upper inequality follows from $e^z \geq 1+z$. For $0 \leq z < 2$ the lower one follows by integrating $[\log((1+z/2)/(1-z/2))]' = 1/(1-z^2/4) \geq 1$; for $z \geq 2$ its left side is nonpositive. Use the upper exponential bound if $E \geq w$ and the lower one if $E \leq w$. The identity

$$U_1 - U_2 = \frac{hz(\lambda E - \delta)}{2(1+z)(1+z/2)} \tag{25}$$

shows that the maximum selects the safe expression. The scalar comparison is monotone within the slab, yielding the whole-slab maximum. In particular U_1 alone is invalid at $E = 0$, $\delta > 0$, $h > 0$.

The independently valid inherited alternative is $E_+ = E/(1+\lambda h) + h\delta$, $E_{\text{whole}} = E + h\delta$. The two recipes are separate native comparison arms. The final algebraic radius is $(8/7)(E_{\text{whole}} + D_g)$. Rational constants, coefficients and radius updates are evaluated outward under the core arithmetic assumptions; nonfinite inputs and failed arithmetic are refused. The dyadic physical clock has no uncharged time-label displacement. Admission compares with the exact rational cap, rather than enlarging it by binary64 rounding.

III. TYPED QUOTIENT OBSERVABLES THROUGH A DISCRIMINANT

a. Theorem T1: fixed rank with multiplicity. For a monic degree-d polynomial P and a polynomial observable h, the multiplication operator on $\mathcal{A} = \mathbb{C}[z]/(P)$ satisfies

$$\operatorname{Tr}_{\mathcal{A}}(m_h) = \sum_{r:P(r)=0} m_r h(r), \qquad \sum_r m_r = d. \tag{26}$$

No square-free condition is needed. Monic division gives a fixed free basis $1, z, \ldots, z^{d-1}$ even before parameter specialization. In the local factor $\mathbb{C}[z]/((z-r)^{m_r})$, multiplication by h is triangular with diagonal $h(r)$ in the basis of powers of $z-r$. Its trace is $m_r h(r)$; summing local factors proves the formula. If the coefficients of P and h depend polynomially on the parameters, polynomial parameter dependence follows directly from monic reduction. This is the classical trace/power-sum relation [2], specialized here with explicit query semantics and finite-arithmetic obligations.

For the original two-equation family $q_1^2 + q_2 = a$, $q_2^2 + q_1 = b$, elimination without division gives

$$P_{a,b}(z) = z^4 - 2az^2 + z + a^2 - b, \qquad q_1 = z, \quad q_2 = a - z^2. \tag{27}$$

The quotient is therefore free of rank four at every (a, b). Its power sums are

$$(S_0, S_1, S_2, S_3) = (4, 0, 4a, -3), \tag{28}$$

$$S_4 = 4a^2 + 4b, \quad S_5 = -10a, \quad S_6 = 4a^3 + 12ab + 3. \tag{29}$$

They follow from Newton identities, or directly by monic reduction and Eq. (26). For $k > 4$, $S_k = 2aS_{k-2} - S_{k-3} - (a^2 - b)S_{k-4}$. In particular $\operatorname{Tr} q_1 = \operatorname{Tr} q_2 = 0$, $\operatorname{Tr}(q_1 q_2) = 3$, and $\operatorname{Tr}(q_1^2 q_2^2) = 4ab + 3$.

At $a = b = 3/4$,

$$P = (z - 1/2)^3(z + 3/2), \qquad S_k = 3(1/2)^k + (-3/2)^k. \tag{30}$$

The triple point contributes length three. Deduplicating roots changes S_0 from 4 to 2 and S_1 from 0 to -1. The accepted query explicitly requests all complex roots with algebraic multiplicity. A selected real branch is a different target: at $(a, b) = (2, 2)$ the regular branch $(q_1, q_2) = (1, 1)$ has determinant $4q_1q_2 - 1 = 3$ and $q_1 = 1$, whereas both $\operatorname{Tr} q_1$ and its average are zero. Thus a trace cannot be substituted as that branch's velocity, even away from a singularity. The native interface also refuses real-only semantics.

A. Shared field and a proved physical contraction constant

The declared aggregate field is

$$\mathcal{F}_1 = -a + S_2/32 + S_4/128 + S_6/512, \tag{31}$$
$$\mathcal{F}_2 = -b + S_2/64 + S_4/64 + S_6/256. \tag{32}$$

Exact substitution yields a single shared nonlinear circuit,

$$N(a, b) = \frac{4a^3 + 16a^2 + 12ab + 16b + 3}{512}, \tag{33}$$
$$\mathcal{F}_1 = -7a/8 + N, \qquad \mathcal{F}_2 = -b + a/16 + 2N. \tag{34}$$

Consequently $\mathcal{F}_2 - 2\mathcal{F}_1 = 29a/16 - b$. An evaluation error e_N in the shared circuit contributes exactly $(e_N, 2e_N)$ before surrounding-operation errors. Outward interval evaluation encloses the original polynomial for the declared parameter intervals and preserves the same query. No inverse root separation enters.

a. Proposition T3: sharp unweighted row-bound constant. On $D = [1/2, 1]^2$, the off-diagonal field derivatives are nonnegative and the diagonals negative. Their row sums increase with a, b. At $(1, 1)$,

$$D\mathcal{F} = \begin{pmatrix} -49/64 & 7/128 \\ 9/32 & -57/64 \end{pmatrix}. \tag{35}$$

The row sums are $-91/128$ and $-39/64$. Therefore

$$\mu_\infty(D\mathcal{F}) \le -39/64. \tag{36}$$

The constant is sharp for this row-sum proof on this domain, since the second row attains equality. Differentiation supplies an offline proof of the bound; the runtime field uses only the polynomial circuit.

The controlled original target used to cross the discriminant is

$$p(t) = \frac{3}{4} + \frac{t}{8} + \frac{t^5}{64}, \qquad -1 \le t \le 1, \tag{37}$$
$$\dot{y} = \mathcal{F}(y) + p'(t)\mathbf{1} - \mathcal{F}(p(t)\mathbf{1}), \tag{38}$$
$$y(-1) = (39/64)\mathbf{1}. \tag{39}$$

Its exact solution is $y = p(t)\mathbf{1}$, with terminal value $(57/64)\mathbf{1}$. Since $p' > 0$ and $\operatorname{disc} P_{a,a} = (4a - 3)^3(4a + 1)$, it crosses the triple-root parameter at $t = 0$. The time-only forcing preserves Eq. (36). A candidate's uniform original-field defect δ and a certified domain tube then give

$$e(t_i + s) \le e^{-(39/64)s}E + \frac{1 - e^{-(39/64)s}}{39/64}\delta. \tag{40}$$

A strict boundary margin with a whole-slab error bound closes the usual first-exit argument when domain containment is not already known. Trace arithmetic included in an original-field interval residual is charged once. Endpoint agreement with $p(t)$ is an independent control and does not replace the whole-path residual/tube obligation.

IV. BOUNDARIES OF THE ADMITTED MATHEMATICAL SCOPE

The selected-branch atlas and the multiplicity-preserving trace have different mathematical targets. The two supplied primitive charts share the determinant $4q_1q_2 - 1$; their transition formulas do not establish regular selected-root continuation at determinant zero. The cyclic domain instead has the strictly positive denominator proved above. Trace regularity at Eq. (30) follows from its monic rank-four algebra and does not resolve a selected singular branch.

The separate F5 global incidence counts also remain open. The original 116/58 ordered/unordered totals omit a horizontal contribution 4/2. The resulting 120/60 remains conditional on completeness of the retained charts and exclusions. A reviewed transversality stratum with $D = 4\eta^3 p + 27\eta^2 q + 27pq = 0$ has local length three despite regular carrier data, so reduced point counting alone cannot certify the quotient rank. A future multivariate trace provider must supply its full ideal, localizations, basis or finite-freeness proof, local lengths and quotient semantics. These incomplete global counts do not affect the independent rank-four proof for Eq. (27).

The native components instantiate the bounded contracts proved here; their measured validation is reported separately. Whole-model physical certification for a different application still requires its original closure/ODE, selected observable, domain, physical clock and representation errors. Generic stiff integration, singular-branch continuation, arbitrary sparse quotients and wider floating-point environments retain independent theoretical and implementation obligations. No generic production claim follows solely from the fixed-model certificates.

V. NATIVE IMPLEMENTATION AND INDEPENDENT VERIFICATION

The cyclic module uses the existing outward arithmetic primitives. Prepared endpoints own the represented bits, model epoch, interval slope, original constraint residual and denominator enclosure. A dyadic grid carries the physical slab clock; finite/domain/epoch/clock failures are explicit refusals. The cap is exactly 1/100000000: a floating-point display value is not allowed to enlarge it. The final admission uses an upward-rounded product with 100000000 and compares with one. An interval dense query encloses the exact field-slope Hermite path to which the proof applies.

The trace module has an explicit all-complex algebraic-multiplicity query. Selected-real and real-only queries are refused. It revalidates public interval fields and rejects nonfinite arithmetic. Constant and linear polynomial queries do not compute unused high moments that might overflow; all supplied coefficients are nevertheless validated. Caller-owned counters retain work on failed requests. The shared-field and necessary-moment routes evaluate the same original aggregate target, with the same time-dependent forcing charged to both trajectory routes.

The benchmark scope includes the private-workspace destruction inside the cyclic total timer, and structured failure diagnostics preserve partial work. It excludes the later consumer lifetime of returned output. This distinction prevents allocation cleanup or failed preparation from disappearing merely because a fast phase has finished.

A. Oracle independence and coverage

The exact Python oracle constructs a finite n-term cyclic inverse action, rather than copying the native affine recurrence. It reconstructs Hermite polynomials in the POWER basis from exact dyadic endpoint values and rational original-field slopes, forms the original slow and constraint residuals, and then converts them to Bernstein form. It checks every exported coefficient interval, original-model domain, clock, radius and cap decision. The trace oracle independently uses exact companion-matrix traces, repeated-root multiplicity controls, the original moment field, and exact polynomial whole-slab residual and tube bounds.

TABLE I. Completed validation at the published execution source. The native unit count is restricted to the two added modules.

Check	Completed result
Cyclic / trace native unit tests	6 / 9 PASS
New ignored tests	0
Independent rational assertions	732,947 PASS
Distinct cyclic slabs / arm–slab records	720 / 2,160
Cyclic coefficient intervals	383,040
Trace trajectories / slabs	10 / 992
Registered Wolfram supporting identities	10 / 10 PASS

A corrupted cyclic coefficient interval and a distinct-root count at the triple point are rejected by negative controls.

No sampled supremum is treated as a proof. The Wolfram identities support exact algebra only; the raw response and file digest are stored, and the analytical proofs and floating-point validation remain separate. SciSpace and Exa provided discovery; the cited primary texts were inspected directly. Inaccessible and irrelevant search results were not used as evidence.

VI. REGISTERED MEASUREMENTS AND PHASE PROFILES

The native run started at 10:37:24 UTC on October 28, 2026, after the source readback at 10:37:20 UTC. Rust 1.94.1 used the locked/offline dependencies and the measurement profile (optimization level 3, no LTO, 16 codegen units). Builds, tests and benchmarks were serialized under a shared lock. The host was Linux on an AMD EPYC 9V74 with nine allowed CPUs; no explicit taskset pinning or frequency control was recorded. These are shared-container phase wall times and source work counters, not hardware performance counters.

Each native timing cell has one separate warmup and seven retained measured repetitions with rotating or alternating route order. No slow repetition was removed. The representative statistic is the median; minimum and maximum are retained in the machine analysis. A ratio is baseline median divided by candidate median. Paired-repeat ratios are a distinct descriptive statistic. These repetitions provide neither a statistical holdout nor a universal hardware-performance guarantee. Phase medians need not sum to the total median.

A. Cyclic: matched cap and fixed candidate comparisons

All three arms use identical compiled RK4 endpoint bits at a given grid. The historical arm uses generic power-polynomial residual construction, convolution and Bernstein conversion with the valid legacy radius. Direct/legacy changes the certification circuit and endpoint reuse while retaining that radius. Direct/tight additionally uses Eq. (23). All share the same outward arithmetic; the baseline was not padded with unnecessary endpoint checks.

TABLE II. First passing registered grid at the exact whole-state, whole-path cap 10^{-8}. Times are whole-call medians in milliseconds. The headline always uses the same Direct/tight arm.

n	Historical, $N = 64$	Direct/legacy, $N = 32$	Direct/tight, $N = 32$	Hist./tight
4	3.148950	0.369877	0.361093	8.7206
8	5.880571	0.640684	0.630028	9.3338
16	11.620309	1.232214	1.263291	9.1984

The whole-state bounds for Direct/tight at $N = 32$ are approximately 8.40591×10^{-9}, 8.40984×10^{-9} and 8.40978×10^{-9} for $n = 4, 8, 16$. Historical $N = 64$ bounds are approximately 1.42×10^{-9}. Thus Table II compares costs meeting the same requested cap, not two trajectories with identical realized error. The first passing grid belongs to the preregistered set $\{16, 32, 64, 128\}$ and is not a proof of a globally minimal step count. Of 36 cells, 24 pass and 12 refusals remain. An upper bound failing to certify the cap does not prove an actual violation.

At fixed $N = 64$, Direct/tight whole-call ratios are 4.2534, 4.6292 and 4.7859 for $n = 4, 8, 16$. Direct/legacy ratios are 4.5295, 4.5019 and 4.8700. This fixes the candidate path and isolates the joint circuit/reuse optimization. The larger matched-cap ratios include the additional benefit of a coarser admissible grid. Endpoint interval slope evaluations drop from 128 to 65, with 63 reused endpoints. At the matched cap they drop from 128 to 33, and candidate RHS calls drop from 256 to 128.

At $N = 64$, preparation plus certificate accounts for 98.8–99.3% of historical total time and 95.4–97.0% of Direct/tight time. The dominant cost therefore remains certification. Direct paths eliminate the generic convolution/transform calls, not every scalar multiplication. The tighter radius decreases the bound but has the same first passing grid as Direct/legacy. Several cells have a slower Direct/tight median with overlapping ranges. It is not an independently established uniform speed improvement. Candidate-only fluctuations do not show a new integrator gain, because the candidate algorithm is unchanged.

B. Trace: local savings and an adverse trajectory statistic

Each field block contains 2,064 evaluations on the same retained inputs. Each trajectory block contains 16 RK4 trajectories with $N = 128$. The necessary-moment field uses 13 scalar multiplications and 9 additions; the shared

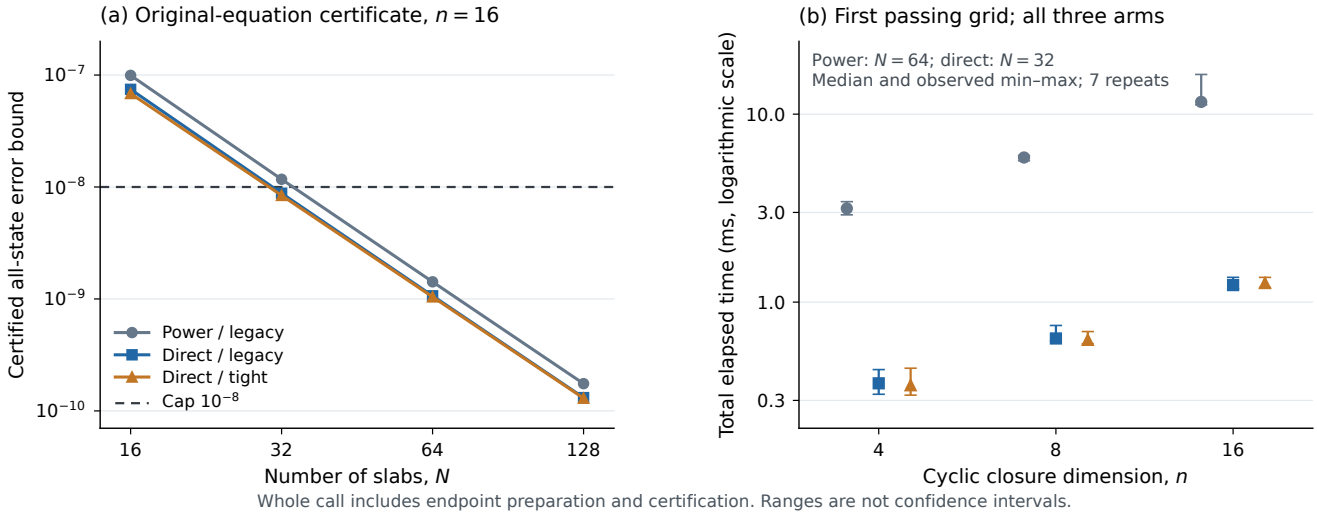


FIG. 1. Cyclic evidence. Left: certified whole-state upper bounds at $n = 16$, not measured true errors. Right: first-passing total medians for all three arms; bars span the retained minimum and maximum of seven repeats. The old and new arms meet the same cap at different registered grids.

field uses 8 and 7. These source arithmetic counters are not measured processor FLOPs. Semantics/finite checks and actual field work are charged; both trajectory routes include the same original forcing.

TABLE III. Trace blocks: seven-repeat medians, with retained min–max ranges in microseconds. Values above one in the last column favor the shared route.

Phase	Direct median [min,max]	Shared median [min,max]	Direct/shared
Candidate field	75.404 [74.502,165.469]	63.145 [62.644,71.337]	1.1941
Interval field	1143.962 [1050.851,1411.494]	763.960 [677.179,1039.664]	1.4974
Combined field	1184.933 [1044.512,1378.796]	926.304 [831.912,1012.564]	1.2792
Candidate trajectory	607.695 [596.027,689.698]	667.394 [577.499,722.538]	0.9105

The shared trajectory marginal median is about 9.824% slower. However, the ranges overlap, the shared route is faster in four of seven paired repeats, and the median paired ratio is 1.00837. We therefore report a mixed descriptive result, not a stable improvement or a universal regression. Field-only savings are real within this cohort but cannot substitute for the complete trajectory result. Driver/forcing/allocation costs require their own measured decomposition before causal attribution.

TABLE IV. Trace accuracy for either route. The official admission remains the preregistered residual certificate. The exact-reference path bound is a separate manufactured diagnostic.

N	Node error	Residual whole-path bound	Exact-reference path bound	Residual gate
16	$5.03622\,10^{-7}$	$2.48785\,10^{-5}$	$1.80247\,10^{-6}$	Refused
32	$3.17551\,10^{-8}$	$2.90805\,10^{-6}$	$1.15923\,10^{-7}$	Refused
64	$1.99206\,10^{-9}$	$3.50361\,10^{-7}$	$7.34774\,10^{-9}$	Refused
128	$1.24713\,10^{-10}$	$4.29562\,10^{-8}$	$4.62445\,10^{-10}$	Refused
256	$7.80109\,10^{-12}$	$5.31653\,10^{-9}$	$2.90033\,10^{-11}$	PASS

The two routes export identical state and time bits on the registered grids. The residual gate first passes at $N = 256$, with bound $5.316525230098761 \times 10^{-9}$. The oracle uses the legacy comparison recurrence, not the tighter rational alternative derived in the theory. At $N = 64$ and 128 the exact-reference whole-path bound already meets the cap. Those residual refusals reveal conservative certification, not a proof that physical error exceeds the cap. We do not change the admission rule post hoc.

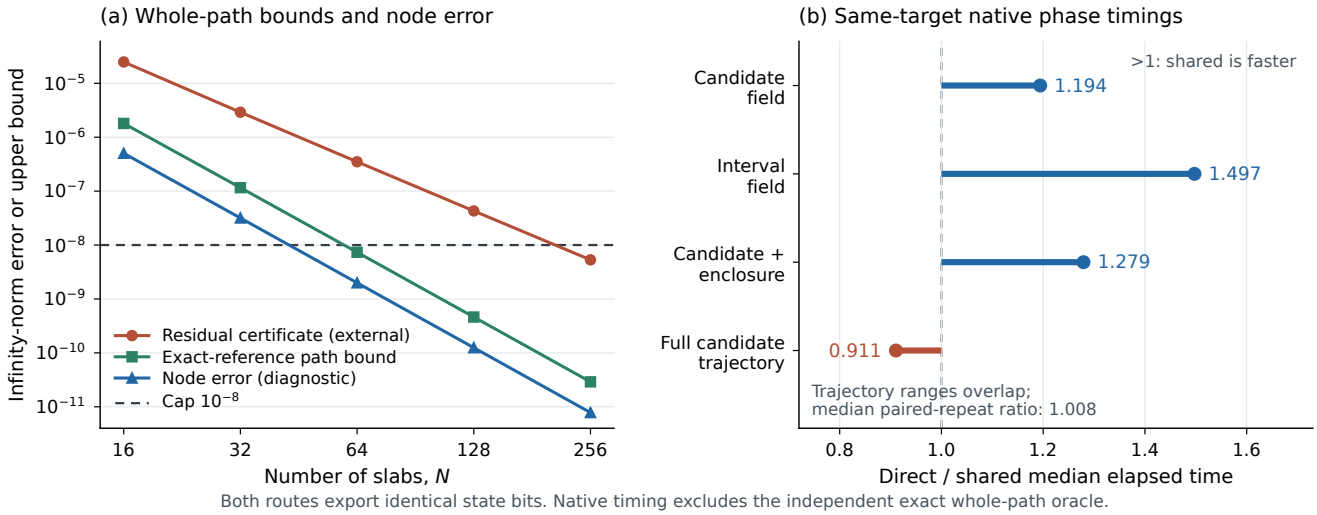


FIG. 2. Trace evidence. The continuous residual certificate is more conservative than the exact manufactured-reference bound. Field-level median ratios exceed one, while the candidate-trajectory marginal ratio does not. The latter has overlapping repeated ranges and is not a stable speed claim.

C. Where certification cost is paid

The external exact oracle internally records 10.775910623 seconds for closure, 2.564409354 seconds for trace and 13.362724506 seconds in total; process wall time is 13.466613229 seconds. No per-route/per-grid trace oracle timings were recorded. Apportioning this aggregate or substituting native point enclosure time would fabricate a complete certified-trajectory ratio.

Closure already returns a native whole-path physical certificate, so that cost belongs to its measured native total; the external oracle validates it independently. Trace has native point/parameter-box enclosures but its whole-path certificate in this experiment is external exact-rational code. No native root-enumeration baseline was executed, and no speed claim against root enumeration is made. Compilation, JSON output, independent validation and consumer output lifetime remain explicitly outside native timing scopes.

VII. POLYALGORITHM INTEGRATION AND REMAINING THEORETICAL WORK

For a common accepted target, complete cost includes session/adapter work, candidate generation, certification and state commitment. Endpoint preparation, failed probes, calibration, rollback and parallel workspace must be charged once. A cheap candidate may lose after its certificate is included. The present data concern structured provider/certificate costs and do not measure the default switcher or mode-to-mode residence decisions.

A. An exact parallel formulation with a serial baseline

The cyclic relation has affine maps $T_i(z) = a_i + c_i z$ with $a_i = f_i$, $c_i = -b_i$. Composition on pairs is

$$(a, c) \circ (d, e) = (a + cd, ce). \tag{41}$$

Expanding both parenthesizations proves associativity. A cyclic composition is $A + Cz$, so its fixed point is $A/(1-C)$; $|C| \le 8^{-n} < 1$ on the box. A balanced prefix/suffix scan has $O(n)$ exact-arithmetic work and $O(\log n)$ span. This does not parallelize arbitrary nonlinear RK stages. The native implementation remains the low-storage $O(n)$ serial recurrence for $n = 4, 8, 16$. A future scan must certify reassociation roundoff or its original tangent residual and include synchronization/workspace costs before claiming a crossover. No new parallel hardware speedup was measured here.

B. Polynomial alternatives and the actual-model remainder

Earlier Chebyshev, Laguerre, Leja, finite-causal and quotient constructions remain conditional routes. A symmetric real spectral interval can justify Chebyshev minimax reasoning; eigenvalues alone do not bound a nonnormal matrix function. Laguerre/Leja actions need a valid domain, tail/degree and rounding bound plus an original-action certificate. A finite structural identity removes iteration only in its declared algebra. A small projected residual does not establish a whole ODE physical-error bound. Selective supplied JVP/AD remains an explicit option when a comparable derivative-free route is insufficient or costs more; no derivative callback is silently activated by the present modules.

The pinned PR95 chemical theory has the frozen structure

$$F(z) = b + Pz + (e^T z)Qz, \tag{42}$$

$$J(z) = P + (e^T z)Q + (Qz)e^T, \tag{43}$$

$$F(z+d) - F(z) - J(z)d = (e^T d)Qd. \tag{44}$$

It can supply a verified $\Gamma\rho^2$ remainder to a near-affine causal certificate. A different carrier J_0 also needs $\|J(z) - J_0\|\rho$. Omitting electron feedback changes the original model. Rate/time enclosures, omitted geometry, full-state norm and lift, and callback rounding remain unclosed obligations. A conditional six-variable identity and entrywise extremizers are not a full-model native qualification.

C. A concrete next certificate question

The gap in Table IV motivates further theory, not a smaller empirical safety factor. A supremum-based defect bound can discard signed cancellation; this is a possible mechanism, not an isolated measured cause of the observed gap. A proposed route is the residual primitive $\rho(t) = \int_{t_i}^{t} R(s)\,ds$ and corrected path $z = \widehat{y} - \rho$. Then

$$z' - F(t,z) = F(t,\widehat{y}) - F(t,z). \tag{45}$$

On the declared box the absolute infinity Lipschitz bound is $L = 75/64$, and the contraction rate remains $39/64$. Conditional on both paths and the relevant tube remaining in that domain, the comparison bound returns to the original reconstruction with an explicit $\|\rho(t)\|$ term and an integral of $L\|\rho\|$. Exact primitive representation, finite arithmetic, domain containment and complete cost must all be established. This is a future task; no implementation or experimental success for it is included in this cohort.

VIII. PRODUCTION QUALIFICATION AND LOCAL HANDOFF

TABLE V. Separate theory and engineering obligations. Current component success does not discharge these gates.

Task	Mathematical obligation	Implementation/validation obligation
Selected singular branch	Physical selection, uniqueness and continuation at a singular point	A typed provider and original-model negative controls
Global F5 / generic quotient	Complete atlas, exclusions, stabilizers, local lengths, finite-free basis and denominator units	Actual multivariate provider and multiplicity checks
Trace path certificate	Sharper valid continuous defect/tube bound	Native outward implementation and route-specific total cost
Actual chemical client	Full rates/time/geometry/norm/rounding remainder	Source-bound provider and external-client validation
Parallel/polynomial routes	Work/span, admissible action domain and finite-error bounds	Hardware crossover with all overhead and failed probes
Adapter and release	Common target and continuation/rollback ownership	Ten interactions, Q-API, authority, supported features and package qualification

Initial results delivery PR97 also exposed the existing comparator build configuration: GCC could not find the MPI header required by its installed SUNDIALS headers. The relevant build/bridge/workflow paths are unchanged from

PR96 and require a supported local compiler/include/link setup. Its fuller CI checkout reported 56 missing authority objects; the separate partial-clone count above has a different object-availability scope. Current delivery state and the focused rustdoc-build receipt are recorded in `CURRENT_STATE.json` and `CI_ADDENDUM_KO.md`.

The unchanged Givens differential test also failed a historical outcome pin: at RHS scale 10^{14} the candidate returned `Ok(3640)` where the recorded table expected an error. Its independent binary64 returned-residual check preceded the failed status assertion. This does not establish an exact rational oracle pass. The actual matrix/RHS bits, profile and platform must be compared before deciding how to separate a historical observation pin from the universal returned-residual contract; no expected flag was relaxed. The earlier scoped Q-GIVENS evidence and this new qualification HOLD coexist.

The machine plan retains the original 23 nodes and 34 dependency edges, with two bounded leaves and current source-bound results. It does not reset correction budgets or mark the six parent N tasks complete. It is a DAG document, not the old execution-suites parser input. The standalone Korean Codex handoff specifies the actual eight-suite plan, exact commands, the required Givens oracle environment and lock ownership.

The local sequence is: (LOCAL-01) establish the newly published local source and reconstruct the preserved evidence; (LOCAL-02) integrate still-needed ancestors once in a fresh worktree based on the latest main and complete the cumulative API/interaction gates; (LOCAL-03) complete real-client providers; (LOCAL-04) perform matched-cap hardware and complete-cost profiling; and (LOCAL-05) qualify the supported release after the theory/provider and hardware branches join. Existing valid evidence is reused within its source and scope. Main is not merged directly, and duplicate ancestor cherry-picks and force pushes are excluded.

The runner must receive the actual published local HEAD and its real remote branch; the historical execution pin must not be forged by changing a tracking ref. New scientific code requires a new source publication and receipt. The general solver remains HOLD until authority, API, negative, ignored-supported-feature and external-client requirements are resolved.

ARTIFACT AND REPRODUCTION GUIDE

The principal files under `research/production_rvj_nag_20261028/` are `RESULTS.json`, `REPORT_KO.md`, `LOCAL_CODEX_HANDOFF_KO.md`, `LOCAL_DEVELOPMENT_PLAN.json`, `analysis/BENCHMARK_ANALYSIS.json`, and `evidence/registered01/RUN_RECEIPT.json`. The raw index supplies both compressed-file hashes and original decompressed byte hashes. Complete repeats, refusals, source identities, actual argv, compiler, binary hashes and timestamps are retained. The figure script only reads frozen analysis; rebuilding this report does not rerun the solver. The repository ledger binds compact numeric results and raw evidence.

[1] V. Magron, A. Rocca, and T. Dang, *Certified Roundoff Error Bounds using Bernstein Expansions and Sparse Krivine–Stengle Representations*, arXiv:1802.04385v1 (2018), https://arxiv.org/html/1802.04385v1.

[2] A. Bostan, P. Flajolet, B. Salvy, and E. Schost, *Fast Computation With Two Algebraic Numbers* (2002), https://algo.inria.fr/flajolet/Publications/BoFlSaSc02.pdf.

[3] J. H. Chaudhry, O. L. Lewis, and M. A. A. Molla, *Adjoint-based A Posteriori Error Analysis for Semi-explicit Index-1 and Hessenberg Index-2 Differential-Algebraic Equations*, arXiv:2507.03712v2 (2025), https://arxiv.org/html/2507.03712v2.